

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 15: Similarity (With Applications to Maps and Models)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two triangles have sides 4 cm, 6 cm, 8 cm and 6 cm, 9 cm, 12 cm. Are they similar? Name the condition.
2. $\triangle ABC \sim \triangle PQR$ with $AB = 5$ cm, $BC = 7$ cm, $PQ = 7.5$ cm. Find QR .
3. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$ cm, $DB = 6$ cm and $AE = 5$ cm. Find EC .
4. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $AB = 6$ cm and $DE = 3$ cm. Find BC .
5. The corresponding sides of two similar triangles are in the ratio 3 : 5. Find the ratio of their areas and of their perimeters.
6. The areas of two similar triangles are 49 cm^2 and 100 cm^2 . If an altitude of the first is 3.5 cm, find the corresponding altitude of the second.
7. In $\triangle ABC$, D and E are the mid-points of AB and AC . Find the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$.
8. A model of a building is made to a scale of 1 : 50. The model is 60 cm high. Find the height of the building.
9. For the model in the previous question, the floor area of a room in the model is 200 cm^2 . Find the actual floor area in m^2 .
10. A map has a scale of 1 : 50,000. Two towns are 8 cm apart on the map. Find the actual distance in km, and the actual area in km^2 of a lake that covers 2 cm^2 on the map.

Answer Key

1. Yes, SSS: each ratio is $\frac{2}{3}$
2. 10.5 cm
3. 7.5 cm
4. 9 cm
5. Areas 9 : 25; perimeters 3 : 5
6. 5 cm
7. 1 : 4
8. 30 m
9. $200 \times 2500 = 500,000 \text{ cm}^2 = 50 \text{ m}^2$
10. 4 km; $2 \times 0.25 = 0.5 \text{ km}^2$